A075 v0.1.0
Hydrodynamische resonante invanging
Falsification protocol and execution report

SST-Workbench | Framework 1.0.10

27 October 2026 | Development-only

1 Identification and status

Catalog A075; family `hydrodynamische_resonante_invanging`; version v0.1.0. The frozen framework is the unmodified shared v1.0.10 runtime, loaded through `framework_bootstrap.py`. Results are a *regularized filament-model screening*, not a qualified full three-dimensional (3D) Euler solution or photon model.

2 Question and objective

Can two or more completely coupled, closed vorticity filaments sustain a lasting excess of incoming-excitation energy in a receiver without damping, imposed speed matching, or a preferred azimuthal mode? This first release checks the necessary numerical conservation and far-separation controls and explicitly leaves the central energy-capture claim unresolved.

3 Competing hypotheses

H_0: the short-run change of the receiver self-Hamiltonian is ordinary conservative energy redistribution; a capture fraction is not defined without a resolved local-energy flux and post-encounter window. H_1: a bona fide finite-energy incoming perturbation exhibits reproducible excess local energy at late time with spatial and temporal convergence and a qualified full-fluid mechanism provider. A short-run numerical PASS cannot establish H_1.

4 Assumptions and limitations

Constant-density, incompressible inviscid fluid background; two backreacting closed, co-axial centerlines; smoothing radius $a > 0$; no reconnection, external forcing, imposed speed matching, damping or imposed azimuthal perturbation. Regularized filament kinematics is not the complete 3D Euler partial differential equation with pressure-Poisson closure. Effective finite-core structure is prescribed rather than resolved. No atomic material response, Maxwell coupling or optical absorption is modeled.

5 Symbols and SI dimensional checks

Symbol	Interpretation	SI units
Γ	circulation	$\mathrm{m^2\,s^{-1}}$
a	regularization length	m
ρ_f	effective fluid density	$\mathrm{kg\,m^{-3}}$
H_a	Hamiltonian proxy	J
t	absolute parameter time	s

The induced-velocity kernel has dimensions $\Gamma\ell^2/\ell^3 = \mathrm{m\,s^{-1}}$. The Hamiltonian kernel has $\rho_f\Gamma^2\ell^2/\ell = \mathrm{kg\,m^2\,s^{-2}}$.

6 Equations and measured observables

$$\dot{\mathbf{x}}_i = \frac{1}{4\pi}\sum_j \Gamma_j \frac{\Delta\boldsymbol{\ell}_j \times (\mathbf{x}_i - \mathbf{m}_j)}{(|\mathbf{x}_i - \mathbf{m}_j|^2 + a^2)^{3/2}}, \tag{1}$$

$$H_a = \frac{\rho_f}{8\pi}\sum_{ij} \frac{\Gamma_i\Gamma_j\,\Delta\boldsymbol{\ell}_i \cdot \Delta\boldsymbol{\ell}_j}{\sqrt{|\mathbf{m}_i - \mathbf{m}_j|^2 + a^2}}, \tag{2}$$

$$H_a = H_{RR} + H_{LL} + H_{RL}, \tag{3}$$

$$\epsilon_H = \max_t \frac{|H_a(t) - H_a(0)|}{\max(|H_a(0)|, \epsilon_0)}. \tag{4}$$

Only $H_{RR}(t) - H_{RR}(0)$ is recorded as a *receiver self-energy proxy*; it is not the trapped incoming energy. A qualified f_{ret} additionally requires a physical control-volume energy-flux calculation.

Machine-readable science contract

Research question Can fully coupled, finite-core closed vorticity lines retain an incoming excitation energy for a preregistered long late interval without dissipative terms, imposed velocity matching or selected angular symmetry?

Objective Negative-control-first screening of mutual vortex-ring dynamics. Report resolved energy redistribution and distinguish it from genuine trapping, 3D Euler closure, and photon absorption.

H_0 H0: observed receiver self-energy response does not establish persistent localized excess energy over four interaction time scales with scale-refined convergence; an unresolved run cannot reject H0.

H_1 H1: a fully coupled, well-resolved ring pair exhibits long-lived localized incoming-excitation energy above the preregistered null, across fixed physical times, insensitive to sampling and angular mode selection.

Registered equations

E1: Regularized midpoint Biot-Savart evolution

$$\dot{\mathbf{x}}_i = \frac{1}{4\pi}\sum_j \Gamma_j \frac{\Delta\boldsymbol{\ell}_j \times (\mathbf{x}_i - \mathbf{m}_j)}{(|\mathbf{x}_i - \mathbf{m}_j|^2 + a^2)^{3/2}} \tag{5}$$

Dimensional check: (L^2/T) L L / L^3 = L/T

E2: Regularized filament Hamiltonian proxy

$$H_a = \frac{\rho_f}{8\pi}\sum_{ij} \frac{\Gamma_i\Gamma_j\,\Delta\boldsymbol{\ell}_i \cdot \Delta\boldsymbol{\ell}_j}{\sqrt{|\mathbf{m}_i - \mathbf{m}_j|^2 + a^2}} \tag{6}$$

Dimensional check: (M/L^3)(L^4/T^2)(L^2/L)= ML^2/T^2

E3: Receiver self-energy change proxy

$$\Delta H_R(t) = H_{RR}(t) - H_{RR}(0) \tag{7}$$

Dimensional check: energy minus energy (J)

E4: Relative conservation drift

$$\epsilon_H = \max_t |H_a(t) - H_a(0)| / \max(|H_a(0)|, \epsilon_0) \tag{8}$$

Dimensional check: ratio of energies (1)

E5: Late-window retention ratio (requires qualified flux)

$$f_{ret} = \frac{\langle E_{loc}^{excess} \rangle_{t\in[T_1,T_2]}}{E_{incoming}^{excess}(0)} \tag{9}$$

Dimensional check: energy / energy = 1

Registered code-to-mathematics steps

1. **S1** [G1; equations E1]: Verify frozen protocol and imported reference checksums
2. **S2** [G2; equations E1]: Construct two independent coaxial closed line samples without imposing mode m=5
3. **S3** [G2; equations E1]: Integrate both moving rings with explicit RK4 and the same finite-core regularized kernel; fixed absolute final time
4. **S4** [G2; equations E2, E3, E4]: Measure H_a, H_RR, H_LL, H_RL and energy conservation at each snapshot
5. **S5** [G2; equations E2]: Test far-separated mutual coupling as null
6. **S6** [G2; equations E1, E2]: Probe temporal and spatial refinement separately at same final time
7. **S7** [G3; equations E5]: Reserve long-window retention criterion; do not infer from H_RR
8. **S8** [G4; equations E5]: Hold-out and native/third-party parity are not available and must not produce positive claims

7 Algorithm: equation-to-code mapping

Build two closed rings with $N = 24$ or 32 segments, smoothing radius $a = 0.22$ in normalized SI-compatible units, $\Gamma_R = \Gamma_L = 1\,\mathrm{m^2 s^{-1}}$, $\rho_f = 1\,\mathrm{kg m^{-3}}$. Integrate both centerlines with explicit fourth-order Runge–Kutta to a fixed final time $T = 1\,\mathrm{s}$ at $\Delta t = 0.05$ and $0.025\,\mathrm{s}$. Test H_a and impulse, a far-separation control, and independent spatial/temporal refinement. No late-time capture status is assigned.

8 Source provenance and independence

`source_evidence/A021_README.md` documents permissible mode decomposition and the requirement that a relative periodic orbit leave and return before claiming recurrence. `source_evidence/A032_README.md` specifies the regularized midpoint kernel, matching Hamiltonian proxy, conservation diagnostics, and observation-window caution. These are *method references*, not independent physical confirmations. Both bytes and SHA-256 values are retained and checked at runtime.

9 Preregistered gates and stop conditions

G0 frozen protocol; G1 source-hash parity; G2 combined reference no-loss (energy and impulse drift below 2%), far-separation cross-energy below 3% of near case, temporal difference below 0.01% and spatial difference below 5%; G3 long-time physical capture (unresolved); G4 independent holdout (unresolved, but attempted); G5 native C++ parity (disabled by profile); G6 GPU screening (disabled); G7 multilibrary replication (disabled); G8 full Euler and pressure-energy flux (unresolved); G9 private reveal withheld. Measured statuses never imply physical certification.

10 Numerical convergence strategy

Fixed absolute end time across all resolution ladders; separate Δt and N controls; record H_{RR}, H_{LL} and H_{RL}, total energy and Kelvin impulse. Production needs finite-core width convergence, controlled arclength remeshing, independent initial-state holdouts and sufficiently long observation windows. A short-run PASS is explicitly diagnostic only.

11 Backend qualification

Python with NumPy binary64 is the only executed solver for v0.1.0. The framework's prior C++ or Intel SYCL parity must never be represented as parity of this solver. Complete Euler certification requires a separate provider; G8 remains unresolved.

12 Blinding and reveal integrity

All source-visible method documents are public. Private reveal data, nonce and forbidden terms remain outside public blind archives. The framework freezes a public commitment and emits uniquely identified run artifacts. Physical claims are not determined from concealed target labels. G9 must remain withheld.

13 Statistics and falsification criteria

Null response is quantified by a far-separated two-ring control. The measured self-energy change is reported but cannot substitute for local capture; no capture-statistic null distribution has been qualified. Frozen numerical thresholds are declared above and never adjusted after outcomes. A lasting absorption claim requires direct late-time f_{ret}, independent holdouts, correct control-volume flux, and convergence.

14 Machine-generated results

Automatic run provenance

Field	Recorded value
Run ID	20261027T033537702994Z-b96e4d46
Run phase	BLIND
Protocol bundle SHA-256 (byte exact)	2acecf5610fa6f50daf3f2b08a41453f2ad31b7665b7e8aa 39967252371719e1
Portable protocol diagnostic SHA-256	4bc33364f3171a4f233c81a367829a9faec06327d85e0e80 1d71ee893b578655
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.13.5 (main, Jul 15 2026, 20:25:40) [GCC 14.2.0]
Platform	Linux-6.18.55-x86_64-with-glibc2.41

Science evidence SHA-256	b6540498d1033d34836f6a66b20c3f54edf96366a6000cb9 8b3e339ea08dd0d3
Final output manifest	See OUTPUT_MANIFEST.json (finalized after PDF and report metadata)

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Is the frozen protocol valid?	Framework verified the frozen protocol before science evaluation
G1	PASS	Are the A021 and A032 methodology files byte-qualified?	Byte-level A021/A032 source README SHA-256 admissibility
G2	PASS	Do numerical no-loss, far-separation and short-time refinement controls pass?	Only short-run regularized filament no-loss, weak-coupling and refinement screening; NOT physical capture
G3	UNRESOLVED	Does fully coupled dynamics demonstrate actual long-time local excitation capture?	No incoming finite-energy wavepacket, late post-encounter window, or full control-volume flux; no evidence of capture
G4	UNRESOLVED	Does an independent preregistered held-out initial-state family confirm actual long-time capture?	No independent capture demonstration to replicate; not a positive holdout result
G5	NOT_APPLICABLE	Does a strict native C++ solver certify parity with Python reference?	Disabled by framework profile
G6	NOT_APPLICABLE	Does GPU screening reproduce certified CPU behavior?	Disabled by framework profile
G7	NOT_APPLICABLE	Do independent geometry/source families replicate the capture result?	Disabled by framework profile
G8	UNRESOLVED	Does a full 3D incompressible Euler provider close pressure-Poisson and energy flux?	Full Euler provider and unconstrained azimuthal mode analysis are not implemented; cannot certify capture
G9	DEFERRED	Private reveal if physically and numerically qualified	Reveal is a separate, commitment-verified phase and is not executed during blind runs.

Scientific qualification (separate from measured gate outcomes)

Gate	Evidence qualification	Blocking prerequisites
G0	QUALIFIED	none
G1	QUALIFIED	none
G2	QUALIFIED	none
G3	UNRESOLVED	none
G4	DIAGNOSTIC_ONLY	G3
G5	NOT_APPLICABLE	G4
G6	NOT_APPLICABLE	G5
G7	NOT_APPLICABLE	G6
G8	DIAGNOSTIC_ONLY	G3
G9	DEFERRED	G8

Backend manifest


```
{
  "diagnostics": {
    "held_out_long_time_not_run": true,
    "mutual_null_to_near_ratio": 0.001295866181088046,
    "physical_capture_fraction_not_measured": true,
    "relative_impulse_drift": 0.0006147967763002576,
    "relative_total_energy_drift": 0.00030644516607354557,
    "spatial_refinement_final_energy_difference": 0.00017178936788188575,
    "temporal_refinement_final_energy_difference": 3.020602917015698e-10
  },
  "native_cpp_parity": "NOT_RUN",
  "python_reference": {
    "actual_backend": "python-numpy-fp64",
    "authority": "REDUCED_MODEL_SCREENING",
    "precision": "float64"
  },
  "requested_native_backend": false,
  "schema": "SST-BACKEND-MANIFEST-2",
  "source_hashes": {
    "A021":
"b615595589551dafbcabddc223e554abeb6ab5a02a064fa96522af73d533e3e9",
    "A032":
"e78411ee0a3a0cd770bd8293e117f44b1908bd419877b5163d5df21b71b9da0e"
  },
  "status": "DEVELOPMENT_ONLY",
  "sycl_gpu_parity": "NOT_RUN"
}
```


Source manifest


```
{
  "manifest_sha256":
"8c5549bc0c48a95485d670821cbcd1161dd75f74e7efc25a7a6c1ec5ccff0060",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "A021",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "A021",
      "path": "/mnt/data/A075_build/A075_hydrodynamische_resonante_invanging/
A075-v0.1.0/source_evidence/A021_README.md",
      "provenance_family": "A021",
      "registry_hint": {
        "catalog_id": "A021",
        "hits": []
      },
      "role": "methodological_reference",
      "sha256":
"b615595589551dafbcabddc223e554abeb6ab5a02a064fa96522af73d533e3e9",
      "source_id": "A021",
      "version": "v0.4.0"
    },
    {
      "catalog_id": "A032",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "A032",
      "path": "/mnt/data/A075_build/A075_hydrodynamische_resonante_invanging/
A075-v0.1.0/source_evidence/A032_README.md",
      "provenance_family": "A032",
      "registry_hint": {
        "catalog_id": "A032",
        "hits": []
```

```
      },
      "role": "methodological_reference",
      "sha256":
"e78411ee0a3a0cd770bd8293e117f44b1908bd419877b5163d5df21b71b9da0e",
      "source_id": "A032",
      "version": "v0.1.0"
    },
    {
      "catalog_id": "A075",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "A075",
      "path": "/mnt/data/A075_build/A075_hydrodynamische_resonante_invanging/
A075-v0.1.0/experiment/model.py",
      "provenance_family": "A075",
      "registry_hint": {
        "catalog_id": "A075",
        "hits": []
      },
      "role": "numerical_solver",
      "sha256":
"242062b9480ff396c6a135fca6d0c7403b6a56a22093d6d415eea3ec36faf47a",
      "source_id": "LOCAL_CODE",
      "version": "v0.1.0"
    }
  ],
  "workbench_root":
"/mnt/data/A075_build/A075_hydrodynamische_resonante_invanging/A075-v0.1.0"
}
```


15 Interpretation and forbidden promotions

A successfully conserved regularized filament Hamiltonian is necessary numerical screening only, not evidence of physical absorption. A fivefold azimuthal mode is not presupposed. Incompressible Euler permits no unmodeled reconnection. This report does not establish an atomic clock, photon identity, or stable quantum excitation.

16 Reproducibility, runtime commands and source code

Use `run_all.cmd FREEZE`, `run_all.cmd BASIC`, then `run_all.cmd FULL` with the pinned canonical Framework v1.0.10. The standard output root is instance-local and uniquely versioned; previous runs are archived. Generated source and output manifests contain SHA-256 checksums. Local offline command: `python -m pytest -q tests` (environment needs NumPy and pytest).

References

[1] P. G. Saffman, *Vortex Dynamics*, Cambridge University Press (1992).

[2] SST-Workbench A021-v0.4.0, *Trefoil Lobe Self-Confinement*, README and method notes (internal source).

[3] SST-Workbench A032-v0.1.0, *Kelvin–Joule Transient Energy*, README and method notes (internal source).